

Proposal for a Windows and Mac Desktop App That Automatically Cleans Spoken Audio in Video Files for YouTube Creators

Executive summary

There is a real, painful workflow gap between “**I recorded a decent tutorial**” and “**this sounds studio quality enough to upload**”. Existing options mostly fall into four buckets: full NLE/DAW workflows such as DaVinci Resolve and RX, which are powerful but slow and cognitively heavy; cloud editors such as Adobe Podcast, Descript, Riverside, VEED, Podcastle, and Cleanvoice, which are fast but upload-dependent and often embedded inside broader editing suites; live-input tools such as OBS filters and NVIDIA Broadcast, which help during capture but are poor fit for fixing already-recorded video files; and pro plugins such as Waves, Accentize, CrumplePop, and Supertone, which assume the user already lives inside an editor or audio host. That leaves a whitespace for a **local-first, file-based, one-purpose desktop utility**: drag in a video, preview “clean speech” vs original, export a remuxed video ready for upload. ¹

The demand signal is credible. Creator forums repeatedly describe current audio cleanup as **resource-heavy, quirky, multi-step, and artifact-prone**. Resolve users complain that Voice Isolation is “resource heavy” and tanks playback, that they spend time “working around all the quirks,” and that exports can differ from timeline playback or fail on certain render modes. OBS users ask for a way to apply noise suppression to recorded files because their current workaround is literally to re-play the audio and re-record it from Windows, and other threads describe the classic tradeoff: background noise decreases, but voices become muffled, robotic, or cut out. ²

The market is not gigantic in the “mass consumer” sense, but it is large enough for a meaningful solo business. A realistic model is not “everyone who uploads video,” but **desktop-first spoken-video creators**: YouTubers, tutorial/screencast creators, course creators, indie commentators, coaches, consultants, and small content agencies. Using creator-economy baselines from SignalFire, Goldman Sachs, YouTube, Microsoft/LinkedIn, and MBO Partners, a defensible planning model yields a **TAM of roughly 2.25 million users, a first-service SAM of roughly 540,000 users, and a 24-month SOM of roughly 4,000 to 10,000 paying users**. At a blended annual ARPU of roughly **\$79–\$149**, this is roughly a **\$178M–\$335M TAM, \$43M–\$80M SAM, and \$0.3M–\$1.5M SOM** revenue opportunity, depending on pricing and execution. These are assumption-heavy planning estimates, not observed market counts, but the order of magnitude is credible. ³

For a solo developer using AI coding agents, this is **buildable but not trivial**. The hard part is not inventing speech enhancement from scratch; strong building blocks already exist. The hard part is productizing them into a workflow that is fast, predictable, cross-platform, and trustworthy on real creator footage. The right target is **not** “match iZotope RX Advanced on every pathological recording.” The right target is: **save creators 10–20 minutes per video, work on ugly but common home-office audio, run locally on Windows and Mac, and export reliable upload-ready files**. If scoped to a file-based desktop tool with excellent A/B preview and guarded presets, an MVP is feasible in about **10–12 weeks** and a polished commercial v1 in about **4–6 months**. ⁴

The strongest recommendation is to build a **Windows/Mac desktop app first**, not iOS first. The problem is born inside a desktop post-production workflow, the media-processing/tooling stack is much better on desktop, and local inference plus FFmpeg-based remuxing are far more natural there. iOS can become a companion later, but it should not be the first wedge.

Problem and customer evidence

What specifically hurts today

The pain is not merely “bad audio.” It is the combination of **time cost**, **tool complexity**, **artifact risk**, and **workflow friction**.

On the Resolve side, creators report that Voice Isolation remains **processor-intensive**, requiring a well-specced machine, and one user explicitly wrote that they “spend so much time working around all the quirks” of the feature’s odd implementation. Other Resolve threads describe export problems: Voice Isolation working only when “Render timeline effects” is checked, skipping/repeating audio on export, or the effect becoming weaker after rendering than it sounded during playback. Those are exactly the kinds of trust-breaking failures that make one-click tools look attractive. ⁵

Reddit creator commentary echoes the same pattern. In one widely indexed Resolve thread, a user described Voice Isolation as “garbage,” saying it was so resource-heavy that “a mere audio plug in” tanked playback. Another reply recommended paying for DxRevive because Resolve’s built-in result could make speech sound “flat or muddy,” while the plugin sounded closer to a studio recording. Even when users think Resolve or similar tools can help, the expectation management is telling: in another thread, a commenter said expecting a “crystal clear” result from damaged audio was unrealistic in any software. That is important from a product standpoint, because it means the winning UX is not magic-marketing but **well-scoped promise plus previewable output**. ⁶

The OBS ecosystem reveals a second pain pattern: creators want capture-time cleanup, but they also want **file-based cleanup after recording**, and today that is awkward. In one official OBS forum thread, a user asked how to use filters on a recorded file because they loved OBS noise suppression but their editor did not have those algorithms; their workaround was to play the audio and record the sound from Windows. Other OBS threads describe the classic suppression tradeoff: lower noise, but voices become muffled, robotic, or aggressively cut off. OBS’s own documentation explicitly says its built-in Noise Suppression filter is effective for **mild** background noise but “not generally effective at large amounts of background noise.” For tutorial creators recording in untreated rooms with fans, keyboards, HVAC, and reverb, that gap matters. ⁷

YouTube itself underscores that poor delivery choices still degrade results. YouTube Help explicitly warns that if audio quality is poor, creators should check whether they delivered a low-quality file such as a low-bitrate MP3, and recommends redelivering an uncompressed audio file where possible. That does not solve room echo or fan noise, but it confirms a product principle: the app should preserve or improve technical delivery quality, not accidentally worsen it through bad defaults. ⁸

Who feels this most acutely

The most promising buyer segments are the ones with **repetitive spoken-video workflows**, modest budgets, and enough output volume that manual cleanup feels expensive.

Persona	Typical workflow	Core pain	Why they would pay
YouTubers and commentators	Talking-head, commentary, educational, review videos	Inconsistent room sound, background noise, multiple takes, upload cadence	Time saved compounds over weekly uploads
Tutorial and screencast creators	Screen recording + voiceover + webcam	Fan noise, keyboard bleed, untreated room, repetitive export flow	Need fast “good enough” every time
Course creators and coaches	Long-form modules, webinars, lesson content	Batch cleanup across many files, low appetite for DAW work	Output consistency matters more than audio geek control
Indie creators and consultants	Solo production, limited gear	USB mic + bad room + no audio expertise	Prefer simple utility over learning Fairlight/RX
Small agencies	Many client footages, repeatable ops	Too many manual cleanup steps per editor	Happy to pay for workflow compression

The surrounding creator-economy data supports this segmentation. MBO Partners estimates **8.1 million US independent creators earning money** in 2023, but most are not rich: **71% earned under \$30,000**, only **9% earned more than \$100,000**, **46% said it is hard to be successful**, and **41% reported burnout**. That combination strongly favors products that save time and cognitive load, but it also means pricing must remain creator-realistic. ⁹

There is also a platform trend tailwind. YouTube says there are **more than 3 million channels in the YouTube Partner Program**, and CEO Neal Mohan said in 2025 that **more than 100 million channels in India uploaded content in the prior year**, highlighting the scale of active creation. LinkedIn is smaller as a creator revenue platform, but strategically relevant: Microsoft said LinkedIn has **1.2 billion members**, that **video is its fastest-growing format**, and that video uploads were up **34% year over year** in FY24 and **more than 20%** in FY25. That makes LinkedIn and Instagram sensible future preset targets, even if YouTube should remain the first beachhead. ¹⁰

What this product should and should not promise

The winning promise is not “salvage any disaster.” It is:

Make spoken audio from normal creator recordings sound cleaner, clearer, more even, and more upload-ready in one pass.

That framing fits both the evidence and the technology reality. You can solve most of the day-to-day creator pain with denoising, gentle dereverb, level consistency, tone shaping, and export hygiene. You should not market it as forensic audio restoration for impossible clips; pro users already have RX, Waves, and Accentize for that. ¹¹

Market sizing

A precise market size for “desktop software that auto-cleans spoken audio in video files” does not exist publicly, so the right approach is a transparent bottom-up estimate.

The broad market backdrop is clearly large. Goldman Sachs estimated the creator economy at roughly **\$250 billion in 2023** and projected roughly **\$480 billion by 2027**. SignalFire estimated roughly **50 million independent creators** globally in 2024. YouTube's paid creator base alone exceeds **3 million** channels inside YPP. ¹²

The narrower opportunity is much smaller than “all creators,” and that is the right way to size it. The product is most useful where all of the following are true: the creator publishes video often, the content is spoken-word heavy, the workflow is desktop-based, and the person is willing to pay to save post time.

Assumptions and sizing model

Layer	Assumption	Calculation	Result
Global creator base	Independent creators globally	Base input	50.0M
Video-first share	Share who regularly publish video	$50.0M \times 40\%$	20.0M
Spoken-word heavy share	Tutorials, education, commentary, podcast-video, interviews, coaching, screencasts	$20.0M \times 30\%$	6.0M
Desktop-first / file-based workflow share	Users likely to value Windows/Mac local processing over mobile-only editing	$6.0M \times 60\%$	3.6M
Frequent-publishing share	Publish often enough that audio cleanup pain is recurring	$3.6M \times 62.5\%$	2.25M TAM users
Initial serviceable scope	English-first, YouTube-first, solo/small-team, local desktop app	$2.25M \times 24\%$	0.54M SAM users
Reachable 24-month share	Solo developer, strong niche GTM, no paid brand budget initially	$0.54M \times 0.7\% - 1.9\%$	4,000–10,000 SOM users

Revenue lens using willingness to pay: if blended annual ARPU is **\$79 low, \$96 base**, or **\$149 high**, the market looks like this:

Market	Users	\$79 ARPU	\$96 ARPU	\$149 ARPU
TAM	2.25M	\$177.8M	\$216.0M	\$335.3M
SAM	540k	\$42.7M	\$51.8M	\$80.5M
SOM	4k–10k	\$316k–\$790k	\$384k–\$960k	\$596k–\$1.49M

These are **planning estimates**, not audited industry counts. The cited external facts are the broad creator-market inputs; the segmentation percentages are judgment calls based on product fit, user pain, and the competitive set. ¹³

Why the pricing band is believable

The current market anchors creator willingness to pay surprisingly well. Adjacent tools sit in a crowded middle range: Descript starts at **\$16 per person per month** on Hobbyist and **\$24** on Creator with Studio Sound included; Riverside's Pro plan is **\$24/month billed annually** or **\$29/month**; Podcastle starts at

\$11.99/month; Cleanvoice starts at **€10** pay-as-you-go for 5 hours; Auphonic offers **2 free hours/month** and then usage-based paid tiers; Resolve Studio costs **\$295 one-time**; Accentize dxRevive is **€99**; RX 12 Elements is **\$99** and RX 12 Standard is **\$399**. A focused desktop utility priced in the **\$99/year, \$12–\$15/month, or \$129–\$149 lifetime-launch** band is therefore inside market norms rather than above them. ¹⁴

Competitive landscape

The market is active, but most products do **not** exactly match the proposed wedge. The whitespace is not “AI audio cleanup” in general; it is **local, one-click, file-based cleanup for spoken audio in video, outside a full editor**.

Competitor comparison

Product	Type	Key features	Pricing	Offline / local	Best fit	Main weakness versus proposed app	Official source
DaVinci Resolve Studio	Desktop NLE	AI voice isolation, music remixer, dialogue separator	\$295 one-time	Yes	Editors already in Resolve	Heavy editor workflow; not a purpose-built pre-upload utility	¹⁵
Adobe Podcast Enhance Speech	Web tool	Speech enhancement, video support, bulk upload on Premium	Free plan + Premium trial shown	No	Fast browser cleanup	Upload-based; part of Adobe web workflow	¹⁶
Descript Studio Sound	Web/hybrid editor	Studio Sound, transcript editing, broader AI editor	Free; Hobbyist \$16; Creator \$24	Mostly cloud workflow	Creators already editing in Descript	Still an editor, not a dedicated ingest-clean-export app	¹⁷
Auphonic	Cloud batch processor	Noise/reverb reduction, leveler, loudness specs, video support	Free 2h/mo; paid usage credits	No	Podcasters, educational content	Usage-metered cloud workflow; less desktop-native	¹⁸

Product	Type	Key features	Pricing	Offline / local	Best fit	Main weakness versus proposed app	Official source
Riverside Magic Audio	Cloud recording/ editor	AI enhancement, reduced noise/reverb, level balancing	Free; Pro \$24 annual / \$29 monthly	No	Recording + editing in one tool	Best inside Riverside ecosystem	19
Podcastle	Cloud creator suite	AI audio/ video editing, noise cancellation, Magic Dust	From \$11.99/mo	No	Broad creator suite users	Broader suite, not narrow desktop utility	20
VEED AI Audio Enhancer	Cloud editor	Remove background noise, “studio-quality” enhancement	Free to try; paid VEED plans	No	Social-video creators in browser	Browser-first editor, not local file utility	21
Cleanvoice AI	Cloud cleaner	Noise reduction, filler removal, mouth sounds, silence	Free trial; PAYG from €10 / 5h	No	Podcasts, talking content	Cloud upload and usage credits	22
Accentize dxRevive	Plugin	Speech restoration with restored missing frequencies	€99	Yes	Post pros and creators using hosts	Requires plugin host/ editor knowledge	23
iZotope RX 12 Elements	Plugin suite	Repair Assistant, voice de-noise, de-reverb, de-click	\$99	Yes	Budget-conscious serious creators	Audio-tool mindset; more manual than one-click utility	24
iZotope RX 12 Standard	Plugin suite + app	Larger restoration toolkit, standalone editor	\$399	Yes	Prosumer / pro post	Expensive and overpowered for simple creator workflow	25

Product	Type	Key features	Pricing	Offline / local	Best fit	Main weakness versus proposed app	Official source
Waves Clarity Vx Pro	Plugin	Real-time automatable voice noise reduction	Captured page showed \$199 sale pricing	Yes	Dialogue post inside DAW/NLE	Host/plugin workflow required	26
Boris FX CrumplePop	Plugin + standalone SoundApp	Echo removal, denoise, leveling, standalone app	Free/ subscription/ perpetual options	Yes	Editors needing restoration plugins	Product family rather than simple single-job utility	27
OBS Noise Suppression	Live filter	Mild background-noise removal	Bundled in OBS	Yes	Live streamers / live capture	Not meant for file-based post on recorded video	28
NVIDIA Broadcast	Live device app	Noise removal, room echo removal, Studio Voice	App + RTX GPU required	Yes	RTX Windows users during capture	Live virtual device, not video-file post-processing	29

Where the gap still exists

The table shows a clear opening:

Most **desktop-local** tools are still **plugins or NLE features**, not a simple standalone creator utility. Most **simple AI tools** are still **cloud/browser-first**. Most **live tools** are not useful once the file has already been recorded. That means the proposed product can be meaningfully differentiated if it does four things together:

1. **Accept video files directly**
2. **Run locally on Mac and Windows**
3. **Optimize for spoken audio only**
4. **Export remuxed upload-ready video without forcing a full edit workflow**

That combination is still under-served. 30

Pricing and go-to-market

Recommended pricing strategy for a solo founder

The best commercial structure is a **hybrid local-first freemium/trial model**, not a pure one-time license and not a cloud-credits model.

A sensible packaging structure would be:

Tier	Recommended packaging	Who it serves
Free	Full preview, short export limits, watermark-free but capped usage	Lets creators hear value before paying
Pro Annual	\$99/year	Core business model; strongest value perception
Pro Monthly	\$15/month	Good for occasional users or launch experiments
Founders Lifetime	\$149 for first 300–500 users	Accelerates early cash and social proof

The logic is straightforward. One-time desktop pricing is attractive because creator audiences like simple ownership. But this product's competitive edge will improve over time through **model upgrades, presets, codec support, platform tuning, signing/notarization work, and bug-fix maintenance**, all of which benefit from recurring revenue. Meanwhile, because the processing is local-first, a free tier does not create the cloud-cost disaster that freemium can create for server-heavy AI products. ³¹

Conversion assumptions and revenue scenarios

For planning, the most useful external benchmarks are directional rather than literal. ChartMogul's 2026 conversion report across 200 software products found a **median free-to-paid conversion of 8% across all products**, with freemium and free-trial mechanics behaving differently. First Page Sage's benchmark data suggests **freemium-to-paid around 2.6%–3.7%** and **opt-in free-trial-to-paid around 17%–18%**. For a creator utility, a prudent plan is to assume you will convert **below generic B2B trial benchmarks**, but with healthier conversion from warm communities and creator referrals. ³²

The following scenarios assume organic launch channels, limited paid acquisition, and a solo-developer operation. These are **modelled scenarios**, not benchmark promises.

Pricing model	Conservative assumptions	Optimistic assumptions	12-month cumulative revenue	24-month cumulative revenue
Freemium + \$99/year Pro	Free→paid 2.5%, modest annual-first mix, 900 paid by month 12 and 2,800 by month 24	Free→paid 5%, strong community uptake, 2,400 paid by month 12 and 7,500 by month 24	\$45k	\$210k
One-time \$129 license	700 licenses in year 1; 1,100 additional in year 2	2,000 licenses in year 1; 3,000 additional in year 2	\$90k	\$232k
Free trial + \$15/mo or \$149/year	800 active paid by month 12; 3,200 by month 24	2,500 active paid by month 12; 9,000 by month 24	\$53k	\$317k

Pricing model	Why it can win	Main downside	12-month optimistic	24-month optimistic
Freemium + \$99/year Pro	Easiest to demonstrate before/after value	Slower cash collection	\$120k	\$610k
One-time \$129 license	Fastest cash, simplest value prop	Lower lifetime value, weaker support economics	\$258k	\$645k
Free trial + subscription	Best long-term economics if retention is good	Harder sell to price-sensitive creators	\$165k	\$924k

My recommendation is to **launch with freemium/trial plus annual pricing**, while using a limited lifetime offer as a launch accelerator rather than the permanent core model.

Distribution and launch tactics

Reddit and forums are viable **discovery** channels, but they should not be your primary transaction layer. In other words: **yes for acquisition, no for checkout**.

For direct sales and compliance, use your own site plus a merchant-of-record/payment layer such as Paddle or an equivalent platform, because it centralizes billing, taxes, and subscription management. Paddle explicitly positions itself as handling billing, payments, and compliance for SaaS and apps. ³³

For distribution itself, the app should use:

- signed Windows installer
- Developer ID signing plus Apple notarization for Mac
- direct download from your website
- optional mirrored distribution through marketplaces such as Gumroad/Lemon Squeezy/Paddle-hosted checkout if you want purchase confidence shortcuts

Apple's documentation is explicit that Mac software distributed outside the App Store should be notarized, and Microsoft documents that SmartScreen reputation affects whether users see warning banners for downloaded Windows apps. If you distribute unsigned or poorly signed binaries, channel feedback will be polluted by trust warnings rather than product quality. ³⁴

The highest-leverage launch tactics for a solo founder are:

Channel	What to do	Why it matters
Reddit/forums	Post before/after examples, offer beta codes, ask for failure clips, obey each community's promotion rules	Best source of pain-language and early trust
Product Hunt	Launch after a small beta, with demo video and benchmark clips	Good for early adopters and feedback loops

Channel	What to do	Why it matters
Creator outreach	DM tutorial creators, screencasters, edtech YouTubers, LinkedIn educators	Best-fit audience with visible pain
Affiliate/referral	20%–30% launch affiliate for creator educators and newsletter operators	Good match for creator-tool sales motion
Comparison content	Publish “Resolve vs one-click cleaner,” “OBS cleanup after recording,” etc.	Captures high-intent search traffic

Product Hunt itself explicitly positions launches as access to a global audience of early adopters and a useful way to gather feedback. Reddit’s official rules and help materials emphasize that communities are community-first, with moderator norms and subreddit-specific policies governing what promotion is acceptable. The practical implication is simple: **lead with usefulness, examples, and feedback requests, not drive-by links.** ³⁵

Product, technology, and architecture

Product scope and feature roadmap

The smartest scope is **spoken-audio cleanup for existing video files**, not live streaming, not full multitrack editing, and not an all-purpose AI editor.

MVP

Feature	User story	Acceptance criteria
Drag-and-drop ingest	As a creator, I can drop MP4/MOV/MKV files into the app	File probe succeeds on common creator formats; unsupported files fail with clear message
One-click speech cleanup preset	As a creator, I can choose “Clean speech” and get a better result without audio knowledge	Output preferred over original in internal listening tests on a majority of noisy creator clips
A/B preview	As a creator, I can instantly compare original vs enhanced	Toggle between versions in two clicks or fewer
Export remuxed video	As a creator, I can export an upload-ready video without re-editing	Original video stream preserved; enhanced audio replaced; sync drift under 40 ms
Loudness consistency	As a creator, my spoken audio comes out more even and less fatiguing	Output stays within defined loudness/peak guardrails
Local-only processing	As a creator, my video does not need to be uploaded	Entire cleanup path works offline after install

v1

Feature	Why it matters	Acceptance criteria
Platform presets	YouTube, LinkedIn, Instagram-friendly speech profiles	Users can export with explainable presets, not hidden magic
Batch queue	Course creators and agencies often process many files	Queue survives multiple files and partial failures
Speech/music awareness	Avoid crushing intentional music beds	Non-speech sections are affected less aggressively
Quality meter	Sets realistic expectations before export	App warns when source is too degraded for "studio" result
Dry/wet control	Helps reduce artifacts	User can back off enhancement if voice sounds overprocessed

v2

Feature	Why it matters
Watch folders / auto-process	Strong agency and team use case
Transcript-assisted cleanup	Lets users jump to problem segments
Team presets and shared profiles	Useful for agencies and small studios
Optional cloud "salvage mode"	For difficult dereverb or severe source repair
API / CLI	Opens B2B and workflow automation paths

UX flow

The ideal UX is short enough to explain in one sentence:

Drop video → hear the difference → export.

More concretely:

1. User drags one or more videos into the app
2. The app probes media, extracts the audio, and shows a "recording quality" estimate
3. The default preset runs immediately on a short preview range
4. User toggles Original / Clean / Strong Clean
5. User optionally reduces or increases intensity with one simple slider
6. User exports remixed video using the original picture stream
7. App offers "Open folder" and "Save this preset as default"

That UX is the real product moat. Many competitors can denoise voice. Fewer let non-technical creators trust the output this quickly.

Recommended technology stack

The best fit is **Tauri + Rust core + React/TypeScript frontend + FFmpeg + ONNX Runtime**, using open-source enhancement models for MVP and leaving room for commercial/licensed engines later.

Layer	Recommendation	Why
Desktop shell	Tauri 2	Smaller bundle size, secure foundation, native system webview
UI	React + TypeScript	Fast solo-dev iteration, strong component ecosystem
Core orchestration	Rust	Performance, memory safety, good fit for media pipelines
Media I/O	FFmpeg	Best-in-class decode, transcode, remux coverage
Inference runtime	ONNX Runtime	Cross-platform inference and hardware acceleration options
Enhancement models	DeepFilterNet baseline; RNNoise low-CPU mode	Strong open-source starting point for speech enhancement
Packaging	Signed installers + auto-update	Necessary for trust and support burden reduction

The official Tauri docs emphasize three advantages: secure foundation, smaller bundle size through the system webview, and frontend flexibility. Electron remains a valid option and is faster for some web-heavy teams, but Electron itself says it embeds Chromium and Node.js, which is exactly why Tauri is usually the better fit for a lean media utility. ONNX Runtime's official docs describe it as a cross-platform model accelerator that supports models from major frameworks. ³⁶

Model options and licensing posture

For an MVP, use the **best open-source local models you can legitimately ship**, then reserve commercial engines for later.

DeepFilterNet is a strong candidate because its project describes itself as a **low-complexity full-band speech enhancement framework**, and the project page shows an **Apache-2.0 / MIT** licensing posture, which is commercially comfortable. The accompanying paper reports real-time viability with a real-time factor of **0.19 on a single-threaded notebook CPU** in its demo setting. RNNoise remains valuable as a lighter low-CPU mode, especially for preview responsiveness. ³⁷

For optional future commercial/licensed engines, the landscape includes:

- **Dolby.io** media enhancement APIs for file-based enhancement
 - **Krisp** AI Voice SDKs for noise cancellation/voice isolation
 - **AudioShake** SDK/API options for real-time separation and speech enhancement
- These are useful for later “salvage,” enterprise, or SDK licensing paths, but they are not necessary to prove the customer problem. ³⁸

The biggest licensing/compliance risk is **FFmpeg**, not the enhancement model. FFmpeg's own legal page says most of FFmpeg is under **LGPL**, but optional parts are **GPL**, and enabling certain components can change compliance requirements. The safe commercial posture is to use an LGPL-compliant build and avoid accidentally shipping GPL/nonfree components unless you deliberately choose that route. You also need to remember that codec patents are a separate issue from code license text. ³⁹

Offline versus cloud tradeoffs

A **local-first architecture** is the right default.

Local-first benefits:

- privacy and trust for client or course footage
- no upload friction
- predictable unit economics for a solo founder
- works offline
- aligns with the whitespace in the current market

Cloud benefits:

- easier model rollouts
- easier handling of heavier dereverb/salvage paths
- simpler telemetry and quality experimentation

The best compromise is:

Local-first MVP, optional cloud enhancement tier later.

That keeps the initial product simple, differentiated, and margin-friendly.

Architecture and processing pipeline

```
flowchart LR
    A[User drops video file] --> B[Probe container and codecs]
    B --> C[Extract audio to PCM]
    C --> D[Analyze speech regions and quality]
    D --> E[Enhancement engine]
    E --> E1[Denoise]
    E1 --> E2[Dereverb]
    E2 --> E3[Voice tone shaping]
    E3 --> E4[Loudness and peak normalization]
    E4 --> F[Preview and A/B compare]
    F --> G[Remux enhanced audio with original video]
    G --> H[Export preset for YouTube, LinkedIn, Instagram]
```

Module breakdown and estimated effort

These estimates assume a solo developer using AI agents for scaffolding, test generation, and repetitive code, but still doing architecture, quality listening, and integration decisions by hand.

Module	Responsibility	Estimated hours
Benchmark corpus and test harness	Build evaluation set of creator clips, regression tests, automatic comparisons	40
Media ingest and probe	Read containers, inspect streams, handle edge cases	35
Audio extraction and remux	FFmpeg-based extraction, replacement, sync-safe remuxing	55
Speech analysis	Segment speech/non-speech, source-quality heuristics	35
Model inference integration	Run enhancement models, chunking, overlap, caching, fallback modes	95
DSP finishing chain	EQ, gentle compression, de-esser/tilt, loudness normalization	55
Preview engine	Fast A/B, preview rendering, waveform/transport	65
Desktop UI shell	Drag/drop, queue, presets, progress, settings	70
Batch export and presets	Multi-file queue, presets, error recovery	40
Analytics, licensing, crash reporting	Basic commercial plumbing	35
Cross-platform packaging	Installer scripts, signing, notarization, updates	50
QA and beta fixes	Real-world codec and content debugging	95
Total		670

With AI agents, that total is not absurd. It translates to roughly **12 weeks at 55–60 focused hours/week**, or **16–18 weeks** at a more sustainable solo pace.

Implementation risks and mitigation

Risk	Why it matters	Mitigation
Overprocessed / robotic voices	Fastest way to kill trust	Use conservative default preset and easy dry/wet rollback
Weak dereverb on bad rooms	“Studio quality” promise can overreach	Quality meter + “salvage mode” warning + preview before export
Sync/export bugs	Video creators will not tolerate broken output	Preserve original video stream, test remux heavily, regression suite
Slow processing on low-end machines	Kills usability	Offer fast/standard quality modes; cache model init; chunked processing
Licensing mistakes	Can block distribution	Use documented LGPL-safe FFmpeg build and audit dependencies

Risk	Why it matters	Mitigation
Mac/Windows trust warnings	Hurts downloads and launch conversions	Sign and notarize from alpha stage onward
Channel spam backlash	Can poison launch	Use feedback-first posts, benchmark clips, and opt-in beta threads

Milestone plan and recommendation

Suggested 12-week plan

Period	Deliverables	Success metrics
Weeks 1–2	Creator clip corpus, benchmark harness, media probe/extract/remux CLI	25+ representative clips; remux success on 95% of test files
Weeks 3–4	First enhancement engine with denoise + basic leveling	Internal listening panel prefers output on at least 60% of noisy clips
Weeks 5–6	Tauri desktop shell, drag/drop, preview, export	End-to-end demo from file import to remuxed export
Weeks 7–8	Better presets, artifact controls, quality meter, batch queue	P50 export time under 2× real time on modern laptops
Weeks 9–10	Private beta with 15–25 creators	Median reported time saved at least 10 minutes per video
Weeks 11–12	Payments, signing/notarization, landing page, launch assets, community seeding	10+ testimonials, 5+ creators willing to pay, crash-free sessions >99% on beta

Recommended launch position

The clearest positioning statement is:

Studio-quality voice cleanup for creator videos in one local desktop step. No NLE, no DAW, no upload required.

That message is:

- obviously different from Resolve/RX/plugin workflows
- obviously different from cloud editors
- immediately legible to tutorial and course creators
- credible if the product output is good

Clear recommendation

Build this as a **desktop-first, local-first speech cleanup app** with a narrow wedge:

- **Target user first:** YouTubers, tutorial/screencast creators, course creators, and solo educators
- **Core promise:** drag in a video, improve voice clarity, export upload-ready video

- **Business model:** free preview / capped export + **\$99/year Pro, \$15/month, limited \$149 lifetime launch tier**
- **Technical stack:** Tauri + Rust + FFmpeg + ONNX Runtime + DeepFilterNet-family baseline
- **Scope discipline:** file-based spoken-audio cleanup only for v1

If you keep the scope there, this is a very plausible solo-developer business. If you broaden too early into full editing, live streaming, multitrack mixing, transcription editing, and all-platform creator suite behavior, it becomes much harder and drifts into direct competition with much larger products.

Open questions and limitations

The biggest uncertainties are not whether the problem exists, but:

- how strong an **open-source dereverb** stage is on real untreated creator rooms versus needing a later commercial or cloud-enhanced path
- how much **lifetime-license demand** exists in your exact audience relative to annual subscriptions
- whether your first best-fit segment is more **YouTube tutorial creators** or **course creators/coaches**
- and, transparently, direct **YouTube comment** evidence was harder to cite than Reddit and forum evidence because public search indexing for YouTube comments is limited; the strongest directly citeable user-pain signals came from Reddit, Blackmagic's forum, OBS forums, and official product/help pages. ⁴⁰

¹ ¹⁵ ³⁰ <https://www.blackmagicdesign.com/products/davinciresolve/studio>
<https://www.blackmagicdesign.com/products/davinciresolve/studio>

² ⁶ ⁴⁰ https://www.redditmedia.com/r/davinciresolve/comments/18aq86v/what_happened_to_voice_isolation/
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⁴ ³⁷ <https://github.com/Rikorose/DeepFilterNet>
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⁵ <https://forum.blackmagicdesign.com/viewtopic.php?f=21&start=0&t=204235&uid=53>
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⁷ <https://obsproject.com/forum/threads/use-filters-on-a-recorded-file.147489/>
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